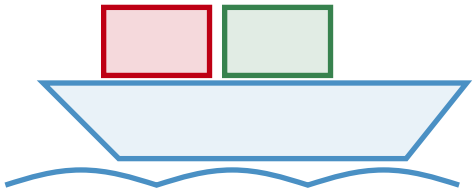
Chapter 9

International Trade



Example: Steel in Isoland

Suppose Isoland initially prohibits international trade in steel. Its domestic price is determined by domestic demand and supply.

If foreign steel becomes available at a lower price, domestic buyers want to purchase it. If foreign buyers offer a higher price, domestic producers want to sell abroad.

Thus, comparing the domestic price without trade with the world price tells us the direction of trade.

The World Price

Definition. The **world price** is the price of a good prevailing in world markets.

A country is a **small economy** in this market if its purchases and sales do not affect the world price. This concerns its influence in the particular market, not its land area or population.

Assume competitive markets, no externalities, identical goods, and no transport costs. With unrestricted trade, the domestic price equals the world price.

Prices and Comparative Advantage

If the domestic price without trade is below the world price, domestic production is relatively inexpensive. The country exports the good.

If the domestic price without trade is above the world price, foreign production is relatively inexpensive. The country imports the good.

In summary, trade patterns reflect comparative advantage. Absolute productivity alone does not determine which goods a country exports.

Example: A Market for Notebooks

Consider a hypothetical small economy with daily notebook demand and supply

$$P = 120 - Q_D, \qquad P = Q_S.$$

Prices are NT$ per notebook. Without trade, $Q_D = Q_S = 60$ and $P = 60$.

Consumer and producer surplus are each $\frac{1}{2}(60)(60) = 1{,}800$, giving total surplus of NT$3,600 per day.

Opening to Imports

Let the world price be NT$40 per notebook. Domestic consumers can buy at this price, and domestic sellers must match it.

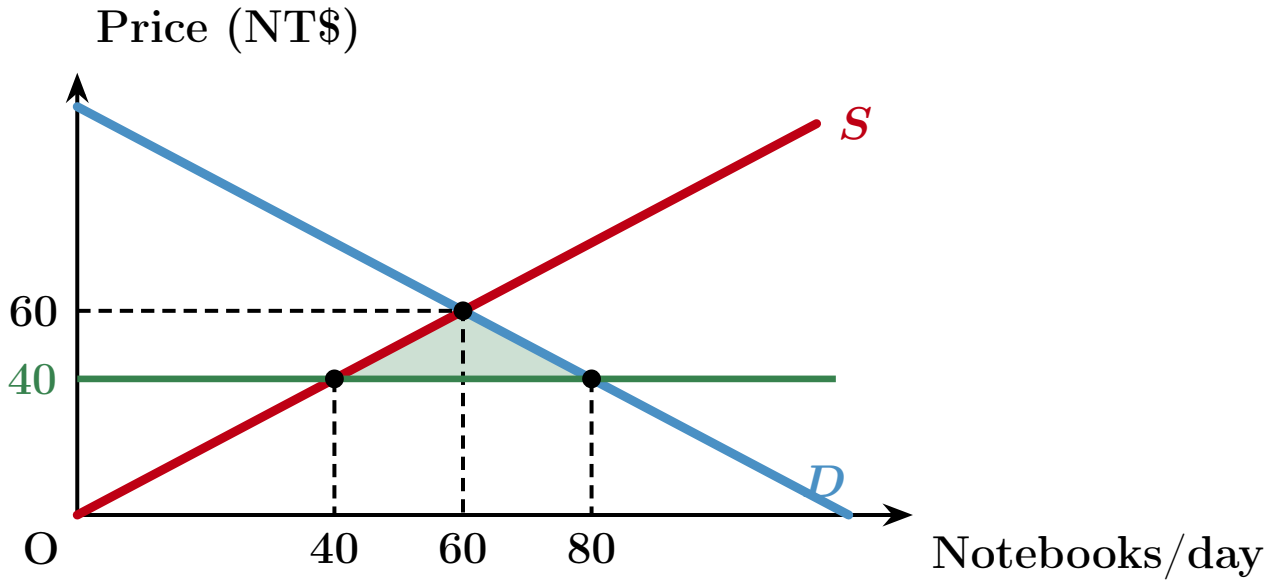
$$Q_D = 120 - 40 = 80, \qquad Q_S = 40.$$

Imports fill the gap between domestic consumption and production.

$$\text{Imports} = Q_D - Q_S = 80 - 40 = 40.$$

Imports in the Market Diagram

The horizontal world-price line determines consumption and domestic production separately.



Domestic production is 40, consumption is 80, and imports are 40 notebooks per day. The shaded area is the gain in total surplus.

Consumers Gain from Imports

The price falls from NT$60 to NT$40, and consumption rises from 60 to 80 notebooks.

$$CS_{\text{trade}} = \tfrac{1}{2}(120 - 40)(80) = 3{,}200.$$

Consumer surplus rises by $3{,}200 - 1{,}800 = 1{,}400$.

The interpretation is that consumers pay less for their original purchases and gain from additional purchases that are now worthwhile.

Domestic Producers Lose from Imports

The lower price reduces domestic production from 60 to 40 notebooks.

$$PS_{\text{trade}} = \tfrac{1}{2}(40)(40) = 800.$$

Producer surplus falls by $1{,}800 - 800 = 1{,}000$.

Overall, domestic producers lose less than consumers gain. Total surplus rises by $1{,}400 - 1{,}000 = 400$.

The Gains from Importing

The economy imports notebooks that would cost more than NT$40 to produce domestically.

It also consumes additional notebooks whose value exceeds the NT$40 world price.

Together, these production and consumption gains equal

$$\tfrac{1}{2}(60 - 40)(80 - 40) = \text{NT\$400 per day.}$$

Example: Wheat Imports in Taiwan

USDA's 2024 report states that Taiwan imports 99% of its wheat demand. Wheat goes into noodles, steamed buns, and dumplings.

In the small-economy model, importing wheat at a price below the no-trade price benefits domestic buyers but reduces domestic wheat growers' surplus.

In other words, the gains come from replacing higher-cost production and making additional purchases worthwhile, the same two sources as in the notebook example.

Source: USDA, Grain and Feed Annual, April 4, 2024, pp. 4–6.

Opening to Exports

Now consider a separate case with a world price of NT$80. Domestic producers can sell abroad at that price.

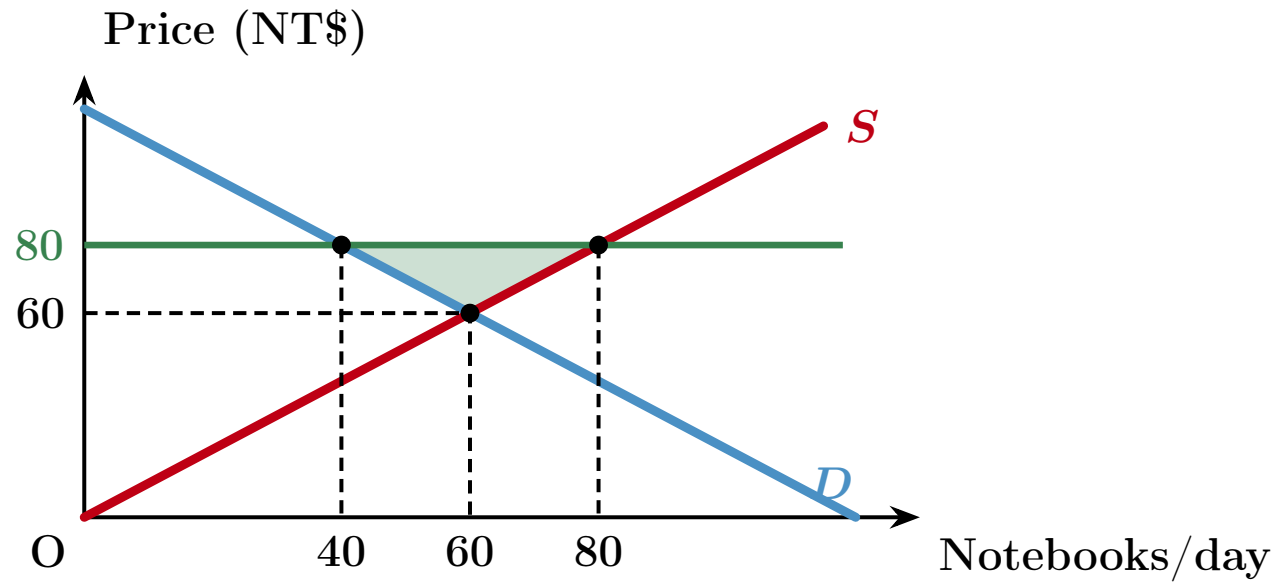
$$Q_D = 120 - 80 = 40, \qquad Q_S = 80.$$

Domestic production exceeds domestic consumption.

$$\text{Exports} = Q_S - Q_D = 80 - 40 = 40.$$

Exports in the Market Diagram

Domestic buyers pay the world price because sellers can obtain that price abroad.



Domestic consumption is 40, production is 80, and exports are 40 notebooks per day.

Welfare in an Exporting Country

At a price of NT$80,

$$CS = \tfrac{1}{2}(120 - 80)(40) = 800,$$

$$PS = \tfrac{1}{2}(80)(80) = 3{,}200.$$

Consumers lose NT$1,000, while producers gain NT$1,400. Total surplus rises by NT$400 per day.

In summary, the winners and losers reverse relative to importing, but the aggregate gain remains positive.

Comparing the Two Directions of Trade

Outcome	No trade	Imports	Exports
Domestic price	60	40	80
Domestic consumption	60	80	40
Domestic production	60	40	80
Consumer surplus	1,800	3,200	800
Producer surplus	1,800	800	3,200
Total surplus	3,600	4,000	4,000

Prices and surplus are in NT$; quantities are notebooks per day, and surplus is per day.

Total Gains Do Not Mean Everyone Gains

In both cases, the winners gain more than the losers lose.

The key distinction is that compensation could make everyone better off, but this does not mean it actually occurs.

In other words, a policy can raise total surplus while imposing substantial losses on particular workers, firms, or communities. Efficiency and distribution are different questions.

Tariffs

Definition. A **tariff** is a tax on goods produced abroad and sold domestically.

For a small importing economy, a tariff t raises the domestic price from P_W to $P_W + t$, provided imports remain positive.

As a result, domestic production expands, consumption contracts, and imports fall. The foreign price stays fixed under the small-economy assumption.

A Tariff on Notebooks

Return to the world price of NT$40. A tariff of NT$10 raises the domestic price to NT$50.

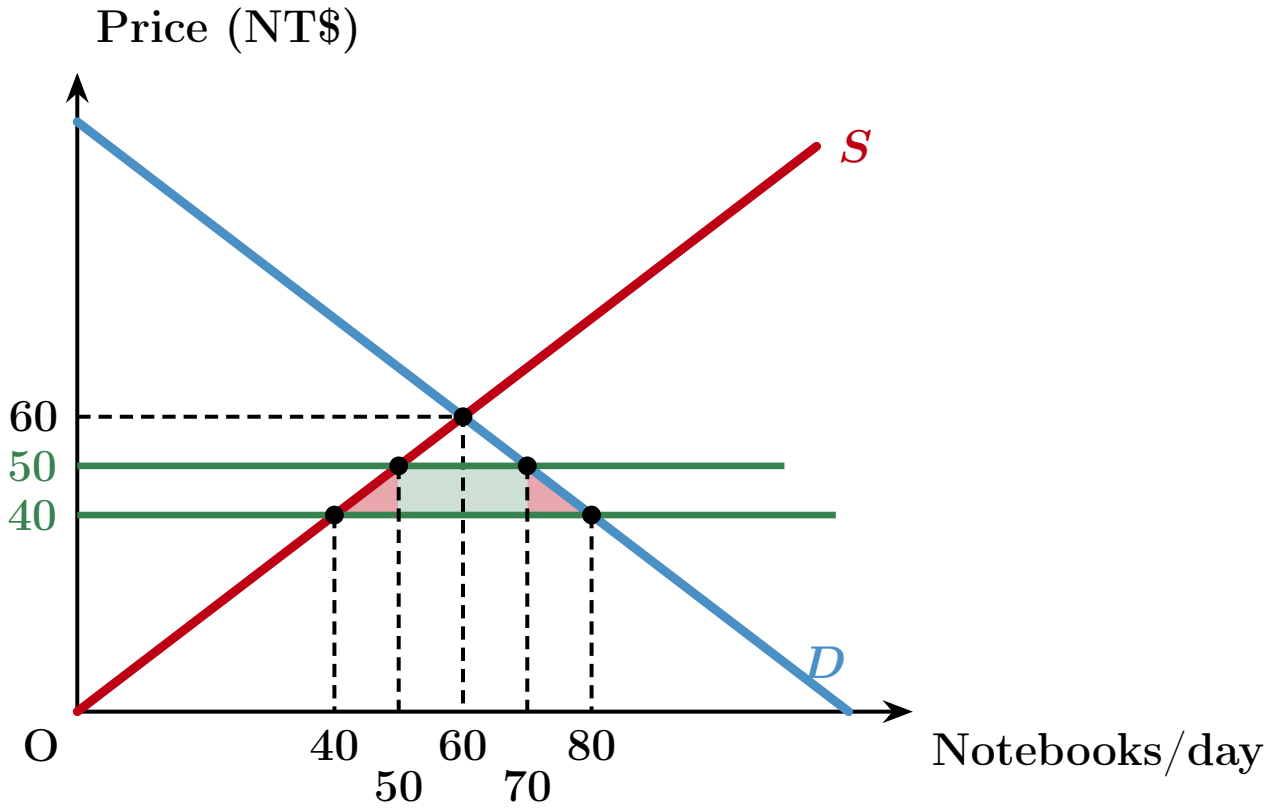
$$Q_S = 50, \qquad Q_D = 120 - 50 = 70.$$

Imports fall from 40 to $70 - 50 = 20$ notebooks per day.

Note that the government taxes imported notebooks, so tariff revenue is $10(20) =$ NT$200 per day.

The Tariff Diagram

The two red triangles show lost surplus. The rectangle between domestic supply and demand at the tariff price is government revenue.



The Welfare Cost of a Tariff

Group	Free trade	With tariff	Change
Consumers	3,200	2,450	−750
Producers	800	1,250	+450
Government	0	200	+200
Total	4,000	3,900	−100

Overall, consumer losses exceed producer gains plus revenue. The difference is the tariff's NT$100 deadweight loss per day.

Two Sources of Deadweight Loss

Production distortion. The extra 10 domestic notebooks cost more to produce than the world price. The loss is $\frac{1}{2}(10)(10) = 50$.

Consumption distortion. The 10 forgone purchases are worth more than the world price but less than the tariff-inclusive price. The loss is also $\frac{1}{2}(10)(10) = 50$.

Together, these losses give $DWL = 50 + 50 = \text{NT\$100}$ per day.

A Prohibitive Tariff

Without trade, domestic supply equals demand at NT$60. This is our no-trade benchmark. With free trade, the actual domestic price is NT$40.

As the tariff rises from NT$0 to NT$20, the domestic price rises from NT$40 to NT$60. At NT$20, imports stop and the no-trade benchmark becomes the actual market price.

Beyond this point, a higher tariff raises the cost of importing, but the domestic price stays at NT$60. For example, a NT$30 tariff makes imports cost NT$70; buyers instead purchase domestically at NT$60.

Thus, at tariffs of NT$20 or more, revenue is zero and the NT$400 gain from trade is lost.

Import Quotas

Definition. An **import quota** is a limit on the quantity of a good that may be imported.

In the notebook example, a quota of 20 imports requires

$$(120 - P) - P = 20,$$

so the domestic price becomes NT$50.

Thus, at this demand and supply, the quota has the same effects on price and quantities as the NT$10 tariff.

Who Receives the Quota Rents?

A permit holder can buy a notebook abroad for NT$40 and sell it domestically for NT$50. The 20 permits create rents of 10(20) = NT$200 per day.

If permits are given to domestic importers, they receive these rents. If the government auctions them competitively, it can collect the rents as revenue.

By contrast, if foreign exporters receive the rents, that income leaves the economy.

A Tariff and a Quota after Demand Rises

With a fixed tariff and continued imports, the domestic price remains P_W+t. Higher demand increases imports.

By contrast, with a binding quota, imports cannot rise. Higher demand instead raises the domestic price and the value of import permits.

In summary, policies with the same outcome at one demand curve need not remain equivalent after demand changes.

Benefits beyond the Surplus Diagram

- Consumers can choose from a wider variety of goods.
- Firms can serve larger markets and spread fixed costs over more output.
- Competition can limit market power and encourage improvements.
- Trade can spread ideas and technology and shift resources toward more productive firms.

The Jobs Argument

Import competition can reduce employment in competing domestic industries. Export opportunities can expand production and employment elsewhere.

Comparative advantage explains why even a country with lower absolute productivity can benefit from trade.

Even so, adjustment can be costly for displaced workers. Protecting one industry's jobs does not by itself establish that a tariff raises total economic welfare.

National Security: Energy in Taiwan

Taiwan relies heavily on imported energy. A disruption to fuel shipments can create costs beyond those in the ordinary market diagram.

Reliable supply can therefore have a national-security benefit. Compare the costs and benefits of protection with alternatives such as fuel reserves, diverse suppliers, and domestic energy sources.

Keep in mind that a tariff raises imported fuel costs without ensuring deliveries. Producers also have an incentive to exaggerate the security benefits of protection.

The Infant-Industry Argument

A new industry may ask for temporary protection while it learns to produce at lower cost.

The government would need to identify industries whose future benefits justify the present costs. Temporary protection can become politically difficult to remove.

However, if future gains belong entirely to the firms, private investors already have an incentive to finance the initial losses.

Example: Learning in Taiwan's Battery Industry

Suppose a new battery producer in Taiwan learns a cheaper production method. Other domestic firms can copy it without paying. This is a hypothetical example.

A **learning spillover** is a benefit from one firm's experience that reaches other firms. The producer does not receive all the benefits, so private investment may be too low.

The policy implication is that assistance may help if its benefits exceed its costs. A tariff also raises buyers' prices; support for training or research may target the spillover more directly.

The Unfair-Competition Argument

Suppose a foreign subsidy lowers the world price of an imported good.

In the small-economy model, domestic consumers gain and domestic producers lose. The consumers' gain exceeds the producers' loss, while foreign taxpayers finance the subsidy.

Thus, a lower foreign price does not by itself justify a tariff on national-welfare grounds. Additional policy concerns must be identified separately.

Protection as a Bargaining Tool

A government may threaten a tariff to persuade another country to remove trade barriers.

If the threat succeeds, both sides may gain from lower barriers. If it fails, the threatening country must either withdraw it or carry out a policy that can harm its own consumers.

In summary, the argument depends on the other government's response, not just the ability to impose a tariff.

Two Approaches to Freer Trade

A **unilateral** approach removes domestic trade barriers without requiring matching changes abroad.

A **multilateral** approach negotiates reductions among several countries. Exporters may support an agreement because it opens foreign markets, balancing pressure from import-competing producers.

The World Trade Organization provides a framework for negotiated trade rules and disputes among members.

Discussion

A small importing economy imposes a tariff. A policymaker says, "Domestic production rises, so the country must be better off."

Explain which groups gain and lose. Identify the missing welfare components in this argument.

Would replacing the tariff with free import permits automatically eliminate the welfare cost?

Answer

Domestic producers gain, but consumers lose from paying more and purchasing less. The government receives revenue.

Consumers' losses exceed producers' gains plus revenue because production shifts toward higher-cost domestic units and some beneficial purchases disappear.

Similarly, an equivalent binding quota has the same distortions. Giving permits away transfers the quota rents to their recipients; it does not remove the lost surplus.

Takeaways

In summary, trade raises total surplus but creates winners and losers.

- The world price determines whether a small economy imports or exports.
- Importing benefits consumers; exporting benefits producers.
- Tariffs distort production and consumption and reduce gains from trade.
- Quota rents depend on who receives the import rights.
- Distribution and adjustment costs matter alongside efficiency.